

Level		OBJECTIVES & REQUIREMENTS	Objective or Requirement	Date Last	Reasoning	Source(s)	System/
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REQ.01.01	System [Mission level]		The multi-agent Europa system shall operate fully autonomously for the entire surface mission duration without requiring real-time command uplinks from Earth.	6/14/2026	The 44-minute one-way light delay between Earth and Jupiter makes real-time demanding physically impossible, all decisions must be onboard. Traceable to OBJ.01	NASA FY 2026-2027 SBIR Solicitation, Topic COMNAV.1. S26B, 2025	System
REQ.01.02	System [Mission level]		The system shall survive and maintain operational status after exposure to total ionizing dose of no less than 100 krad over the nominal mission lifetime.	6/14/2026	Jupiter's magnetosphere delivers radiation doses far exceeding near-Earth missions. Europas Clipper's radiation vault architecture targets ~100 krad as the design threshold. Radiation environment directly drives the shielded vault requirement. Traceable to OBJ.01	National Academies, Origins Worlds and Life Decadal Survey 2023–2032, https://www.nationalacademies.org/read/26522 ; Galileo spacecraft radiation vault design heritage	System; Electronics; Power
REQ.01.03	System [Mission level]		The system shall maintain a minimum inter-agent network connectivity of two active relay nodes at all times during surface operations to ensure consensus is achievable.	6/14/2026	Consensus algorithms require a minimum connected graph to guarantee convergence. Two active relays plus a surface rover guarantee a 3-node connected topology as the minimum viable configuration. Traceable to OBJ.01 and OBJ.04.	NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024, https://www.nasa.gov/wp-content/uploads/2024/07/civil-space-shortfall-ranking-july-2024.pdf	System; COMMs
REQ.02.01	Subsystem [Operations Level] Mobility		The surface robot shall maintain a localization accuracy of no worse than +-0.5 m relative to its last known position during any traverse segments up to 50 m in the absence of GPS.	6/14/2026	Cryobot insertion site selection requires meter-level positioning to match the radar identified thin ice candidate. +-0.5 over 59 m segments in consistent with visual odometry performance on Mars rovers. Traceable to OBJ.02	NASA FY 2026-2027 SBIR Solicitation, Topic COMNAV.1. S26B; Mars Curiosity rover visual odometry heritage	Mobility; GNC
REQ.03.01	Subsystem [Operations Level] Science / Payload		The ground-penetrating radar sounder shall resolve ice shell thickness to within +-10% accuracy for ice layers between 1 km and 30 km depth, producing a subsurface map at the minimum spatial resolution of 100 m along the traverse track.	6/14/2026	Identifying optimal cryobot insertion site requires knowing where the ice shell is the thinnest. +- 10% thickness accuracy is enough to distinguish viable drilling candidates. The 100 m along track resolution is consistent with the Europa Clipper REASON instrument design parameters. Traceable to OBJ.03	National Academies, Origins Worlds and Life Decadal Survey 2023–2032. https://www.nationalacademies.org/read/26522; Europa Clipper REASON instrument documentation	Science/Payload, Power

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REQ.04.01	Subsystems [Operations Level] COMMS		The inter-agent mesh network shall achieve consensus on a shared terrain map update within no more than 10 minutes under a degraded network condition defined as packet loss rates of up to 30% on any single inter-agent link	6/14/2026	Rover traverse planning depends on a current consensus validated terrain map. A 10 minute convergence bound limits traverse planning delay to an acceptable operations cadence. 30% packet loss represents a plausible radiation induced degradation scenario for the environment. Traceable to OBJ.04.	NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024; NASA FY 2026-2027 SBIR Solicitation, Topic COMNAV.1.S26B	COMMs, CDH
REQ.03.02	Subsystem [Operation Level] Science / payload		The radar sounder antenna shall be mounted such that the antenna boresight remains within +-5 of nadir during all active sounding operations; the rover shall halt traversal if the measured surface slope exceeds 20 to maintain pointing compliance.	6/16/2026	A nadir pointing antenna is required to minimize geometric error in the two-way travel time measurement. . Off nadir tilt reduces basal reflection return strength and introduces range error that degrades	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; Europa Clipper REASON instrument documentation	Science/Payload; Structures; Mobility
REQ.03.03	Subsystem [Operations Level] Science / Payload		The rover power subsystem shall reserve a dedicated peak power allocation for radar sounder transmission pulses and shall command drive motor shutdown during each active sounding window to prevent power contention with the RTG continuous output.	6/16/2026	Radar sounders draw high peak pulse power during transmission. The RTG provides fixed continuous output; simultaneous drive and sounding would exceed available power margin and risk under-powering the radar pulse, degrading return signal amplitude below the noise floor. Stop-and-sound is consistent with high-power instrument operation heritage on Mars rovers. Traceable to OBJ.03. At approximately 40 traces per sol across four data sheets, raw science data accumulates faster than the non-deterministic Earth relay can downlink. Seven sols of buffer covers the consensus convergence window and plausible communication gaps. Lossless compression preserves the measurement fidelity required by REQ.03.01 while reducing relay load. Traceable to OBJ.03 and OBJ.04.	Europa Clipper REASON instrument documentation; NASA RTG power system heritage	Science/Payload; Power; Mobility
REQ.03.04	Subsystem [Operations Level] Science / Payload		The command and data handling subsystem shall provide onboard storage sufficient to retain a minimum of 7 sols of uncompressed radar trace data, and shall apply lossless compression to all Radar_Traces records prior to inter-agent relay transmission, achieving a minimum compression ratio of 2:1.	6/16/2026		NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024; Europa Clipper REASON instrument documentation	Science/Payload; COMMs; CDH

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REQ.05.01	Subsystem [Operational Level] Thermal		The rover thermal control system shall maintain all avionics and battery cells within an operational temperature range of 0C to 40C during all active surface operations and above -20C during night-mode low-power thermal management, using active heater elements powered by RTG waste heat.	6/28/2026	Europa's surface temperature is approximately 90K (-183C). Battery cells experience severe capacity loss below -20C and permanent damage below -40C. The RTG provides continuous waste heat that can be routed to a thermal enclosure, making active heating feasible without a dedicated power draw penalty. No ground intervention is possible to recover a thermally failed rover. Traceable to OBJ.01.	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; NASA RTG thermal heritage	Thermal;Power; Structures
REQ.05.02	Subsystem [Operations Level] CDH		The command and data handling subsystem shall detect and autonomously recover from a single-event latchup in any processor core within 330 seconds of occurrence, restoring nominal operations without ground intervention, and shall log each latchup event to non-volatile memory with a timestamp and affected core identifier.	6/28/2026	Jupiter's magnetosphere produces a high-energy particle flux that induces single event latchup at a far higher rate than near-Earth missions. With a 44-minute one-way light delay, no ground intervention can recover a locked core in time to prevent missions tall. The 30-second recovery bound prevents autonomous traverse planning and site selection from stalling indefinitely. Traceable to OBJ.01, linked to RISK-02.	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024; Galileo spacecraft radiation heritage	CDH;Electronics
REQ.05.03	Subsystem [Operations Level] Power		The rover power management subsystem shall enforce a per-sol energy budget such that total energy draw across all subsystems does not exceed 90% of the RTG's nominal daily energy output, reserving a minimum 10\$ margin for autonomous anomaly response, heater activations, and unplanned communication relay passes.	6/28/2026	The RTG provides fixed continuous output with no ability to surge on demand. Radiation-induced matchups, slope-stop halts per REQ.03.02, and degraded consensus requiring extra relay passes all introduce unscheduled power draws. A hard 10% reserve prevents the rover from entering a power fault state during autonomous recovery events with no ground support available. Traceable to OBJ.01 and OBJ.03	NASA RTG power system heritage; Europa Clipper power budget architecture; NASA FY 2026–2027 SBIR Solicitation, Topic COMNAV.1.S26B	Power;CDH;Thermal; Mobility

	Status?		Associated Risks with		Testing Method	Verification Plan	Validation	Any Pre-	Notes For
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REQ.01.01	Unaccomplished		Risk-01: Single-point autonomy failure with no fallback human override		Hardware-in-the-loop simulation. Run a 72-hour autonomous mission sequence with a simulated 44-min communication blackout, and verify zero unresolved command dependencies on Earth.	Verify the software autonomously executes all planned traverses, consensus rounds, and science data collected without any Earth uplink during the blackout window.	Demonstrate that the mission objectives are achieved across three independent HILL runs without Earth support		
REQ.01.02	Unaccomplished		RISK-02: Accumulated single event upsets degrade consensus algorithm convergence before mission completion.		Co-60 gamma irradiation of all radiation-sensitive components to 110 krad (10% margin). Monitor functional performance throughout exposure.	Verify all components pass functional tests at end of life dose with no loss of required functionality	Validate the integrated rover and relay nodes complete simulated mission operations after TID exposure.		
REQ.01.03	Unaccomplished		RISK-03: Simultaneous radiation-induced failure of multiple relay nodes isolating the surface rover.		Network fault simulation: simulate progressive relay node failures and verify consensus detects degenerated topology and triggers reconfiguration before dropping below 2 active nodes.	Verify that fault detection and reconfiguration logic activates within one consensus cycle period upon detection of a node loss event.	Validate that the surface rover completes a traverse plan using map data from a degraded 2-relay configuration.		
REQ.02.01	Unaccomplished		RISK-04: Ice surface texture homogeneity reducing visual odometry feature tracking quality.		Field test on analog terrain: command 50m traverse segments and compare onboard state estimate to ground truth GPSD position across 20 runs under varying terrain conditions.	verify that position error remains within +-0.5 m across a minimum of 20 traverse segments.	Validate that the rover navigates to and stops within +-0.5 m of a designated waypoint representing a candidate insertion site.		
REQ.03.01	Unaccomplished		RISK-05: Ice shell heterogeneity or brine inclusions creating radar clutter that degrades thickness estimates.		Laboratory radar range measurements against ice targets of known thickness using dielectric matches analogs, verify +- 10% accuracy across the 1-30 km depth range.	Verify that the radar sounder produces subsurface profiles with +-10% depth accuracy against calibration targets before rover integration.	Validate that the subsurface map produces during a field traverse correctly identifies presurveyed thin ice candidate sites within the required spatial resolution.		

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REQ.04.01	Unaccomplished		RISK-06: Radiation induced burst errors exceeding 30% packet loss on multiple links simultaneously, stalling consensus.		Software-in-the-loop consensus simulation: Introduce 30% packet loss on each inter-agent link and measure convergence time across 100 test runs.	Verify the consensus convergence time remains <10 minutes across all 100 test runs with 30% packet loss.	Validate that the rover initiates a traverse using a map update produced by the consensus protocol under the degraded link test condition.		
REQ.03.02	Unaccomplished		RISK-05: Ice shell heterogeneity creating radar clutter that degrades thickness estimates; RISK-04: Ice surface texture reducing visual odometry quality (slope-stop logic dependency)		Bench test antenna mounting fixture: tilt rover chassis to 5° increments and measure antenna boresight deviation using an inclinometer. Verify pointing error stays within ±5° across the full range of surface slopes up to 20°.	Verify that antenna boresight deviation does not exceed ±5° of nadir across all chassis tilt angles up to 20° slope	Validate that the rover correctly halts traversal and resumes sounding when slope threshold is exceeded during an analog terrain field test.		
REQ.03.03	Unaccomplished		RISK-02: Accumulated single event upsets degrade system performance; RISK-05: Degraded radar return amplitude from insufficient transmission power		Hardware-in-the-loop power budget simulation: command simultaneous drive and sounding operations and monitor RTG output margin across 50 test cycles. Verify drive shutdown command fires before each sounding pulse.	Verify that drive motors are commanded off prior to every sounding transmission and that RTG output margin remains positive across all test cycles.	Validate that the rover completes a full simulated traverse sol with correct stop-and-sound behavior and no power fault events.		
REQ.03.04	Unaccomplished		RISK-06: Radiation induced burst errors exceeding 30% packet loss stalling consensus; RISK-05: Ice shell heterogeneity generating excess CLUTTER traces inflating data volume		Software-in-the-loop storage simulation: generate 7 sols of synthetic radar trace data at full volume and verify onboard storage does not overflow. Apply compression algorithm and measure output ratio across 100 data sets.	Verify that onboard storage accommodates 7 sols of uncompressed data and that compression ratio meets or exceeds 2:1 across all 100 test data sets.	Validate that compressed science data downlinked through the inter-agent mesh under 30% packet loss conditions can be fully reconstructed without data loss on the receiving node.		

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REQ.01.01

REQ.01.02

REQ.01.03

REQ.02.01

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REQ.04.01

REQ.03.02

REQ.03.03

REQ.03.04

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REQ.05.01

REQ.05.02

REQ.05.03